Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-PICU-POL-5-Vers.2.0

Effective Date: 12/10/2023
Target Review Date: 10/10/2026

Title: Non- invasive ventilation and High Flow Nasal Cannula for Pediatric acute respiratory distress

Non -invasive ventilation (NIV) and high flow nasal cannula (HFNC) are used as modes of respiratory treatment in the pediatric population. They are initiated as first line respiratory support in patients with respiratory distress.

Advantages over invasive ventilation include:

1. Preserving spontaneous respiratory effort
2. Reducing acute lung injury related to invasive ventilation
3. Reducing need for sedation and analgesia
4. Reducing risk of ventilator associated pneumonia

Indications for non -invasive ventilation use:

1.Children presenting with respiratory distress with /without abnormal Blood gas results (PH < 7.3 , Pco2 50-60 mmHg with Spo2 < 92% in RA)

2. Upper airway obstruction such as obstructive sleep apnea or post -extubation stridor (*Not to be used in upper airway emergencies such as croup or tracheitis)*

3. Apnea with Consideration of early mechanical ventilation for patients with recurrent apnea or significant apnea

Contraindications for Use of NIV :

1.Need for immediate airway intubation for peri-arrest situations

2. Significant hemodynamic instability and inotropic need

3. Encephalopathy and inability to control the airway and secretions

4. Recurrent apnea

5. ARDS with severe hypoxia(P:F ratio or SF ratio < 150)

6. Recent abdominal surgery including unrepaired trachea-esophageal fistula and congenital diaphragmatic hernia

Initiation and monitoring of NIV:

The indication for initiation of NIV should be reviewed with the attending consultant.It is advised to start with lower pressure on NIV settings with gradual increase to allow the patient to tolerate it. The patient should be continuously monitored for heart rate , respiratory rate and oxygen saturation.

It is a must that frequent and regular physical assessment are carried for signs of deterioration or improvement. Most of the responders to NIV show response within the initial 2 hours of starting NIV. However, some of the patients may have delayed response. Children with clinical response improve with regard to their respiratory effort, heart rate, oxygen saturation and gas exchange. The pediatric ICU team should be involved in patients with significant respiratory distress on presentation and in patients with no clinical response following the initial 2 hours of treatment or earlier if signs of significant respiratory distress or failure.

For initiation and monitoring see NIV initiation algorithm

NIV interface

The interfaces in use are nasal masks, oro-nasal masks and total face masks. An ideal interface should be comfortable, has minimal air leak around the mask and cause minimal claustrophobia(irritability).

The interface selection should be individualized but, the following suggestions are to be considered:

Nasal masks / nasal prongs : They are used for neonates and small infants. Aim to minimize the leak from mouth with use of pacifier.

Oro-nasal interfaces and total face masks : for older infants and children

NIV devices in use

(expiratory positive pressure) representing positive airway pressure during expiration. There is minimum difference of 4 between IPAP and EPAP to generate effective tidal volumes.

Caring for a child on NIV:

Infants and children on NIV should have continuous monitoring of their vital signs with appropriate nursing ratio to ensure adequate delivery of care.

Medical and nursing care should include the following:

- Prevention of pressure ulcers on the face by using dressings over the pressure points .It is a must that inspection for redness and skin break down is included during nursing care. Skin redness of breakdown need to be reported and indicated in nursing observations chart.
- Use of artificial tears every 4-6 hours when total face masks are used to prevent dryness of the eyes
- Use of nasogastric feeding tube for air decompression

Common Sedation choices :

Sedation and analgesia is used to help the child tolerate NIV and to facilitate synchrony with the machine. Medications doses should be used in amounts that does not interfere with the child's breathing.

Common agents used are:

Chloral hydrate: It is widely used for painless pediatric procedures. The onset of action is within 30 minutes and has duration of 4-6 hours.

Usual doses range from 25 to 50 mg/kg

Dexmedetomidine: is preferred to be used in this context. It has anxiolytic, sedative effects and preserves spontaneous breathing. It requires continuous vital signs monitoring as it can produce adverse hemodynamic effects including bradycardia.

- It is a potent hypnotic agent and analgesic with a dissociative effect. It has minimal effects on respiratory function and produces a bronchodilator effect by relaxing smooth bronchial muscles. It can also, increases production of secretions.

Other agents like morphine can also be used provided that they are used in smaller doses that does not interfere with the child's breathing.

****Please, avoid use of sedation in children with apnea***

Weaning of NIV :

Weaning is initiated once the child's respiratory distress improves(preferably in 24 hours post- initiation) ***see weaning algorithm***

High flow nasal cannula treatment

HFNC delivers a humidified oxygen and gas mixture heated to approximately 34 ° C, allowing higher air flow rates exceeding 2 L/minute. These features enable the delivery of air flow equal to or greater than the inspiratory flow of a spontaneously breathing patient. The use of HFNC has broad indications, including different causes of respiratory distress. Overall, HFNC is most commonly used among children with bronchiolitis and pneumonia. In this institution, we reserve use of HFNC for infants and children with milder respiratory distress.

For initiation and weaning of HFNC, see related algorithms

distress

Are there contra-indications for NIV?

Yes
Consider early invasive ventilation

Contra-indications to NIV

1. Need for immediate airway intubation for peri-arrest situations
2. Significant hemodynamic instability and inotropic need
3. Encephalopathy and inability to control the airway and secretions
4. Recurrent apnea
5. ARDS with severe hypoxia(P:F ratio or SF ratio < 150)
6. Recent abdominal surgery including unrepaired trachea-esophageal fistula and congenital diaphragmatic hernia

No

Indications for non -invasive ventilation use:

1.Children presenting with respiratory distress with /without abnormal Blood gas results (PH < 7.3 , Pco2 50-60 mmHg with Spo2 < 92% in RA)

2. Upper airway obstruction such as obstructive sleep apnea or post -extubation stridor
(Not to be used in upper airway emergencies such as croup or tracheitis)

3. Apnea with Consideration of early mechanical ventilation for patients with recurrent apnea or significant apnea

What type of respiratory failure?

Type 1 respiratory failure with hypoxemia

Initial settings :
CPAP: 6-8 cm H2O
Fio2 50%-60% to target Spo2 > 93%

No clinical improvement
*Change to BiPAP
IPAP: 8 to 10 cm H2O
EPAP/PEEP: 6 to 7 cm H2O
FiO2: 50 to 60%
*Request PICU review and consider invasive ventilation as applicable

*Increase PIP by 2 cms every 5 minutes in the initial 30 minutes to your target PIP

Target:
-PIP range: 10 to 16 cm H2O
Max. PIP 20
-EPAP/PEEP: 6 to 8 cm H2O
-Tidal volume: 8 to 10 mL/kg
-Target reduction FiO2 < 40% within 24 h

Type 11 respiratory failure with Co2 retention

Initial settings :
BIPAP
IPAP: 8 to 10 cm H2O
EPAP/PEEP: 6 to 7 cm H2O
FiO2 : 50%-60% % to target Spo2 > 93%

*Increase PIP by 2 cms every 5 minutes in the initial 30 minutes to your target PIP

Target :
-PIP range : 10 to 16 cm H2O
Max. PIP 20
-EPAP: 5 to 8 cm H2O
-Tidal volume: 8 to 10 mL/kg
-Target reduction FiO2 < 40% within 24 h
-Request PICU review and consider invasive ventilation as applicable

Assess for clinical improvement within 60-90 minutes of starting NIV
look for:
- improvement in clinical work of breathing.
- Decrease in heart rate, respiratory rate by 20% from baseline. And improvement in blood gas

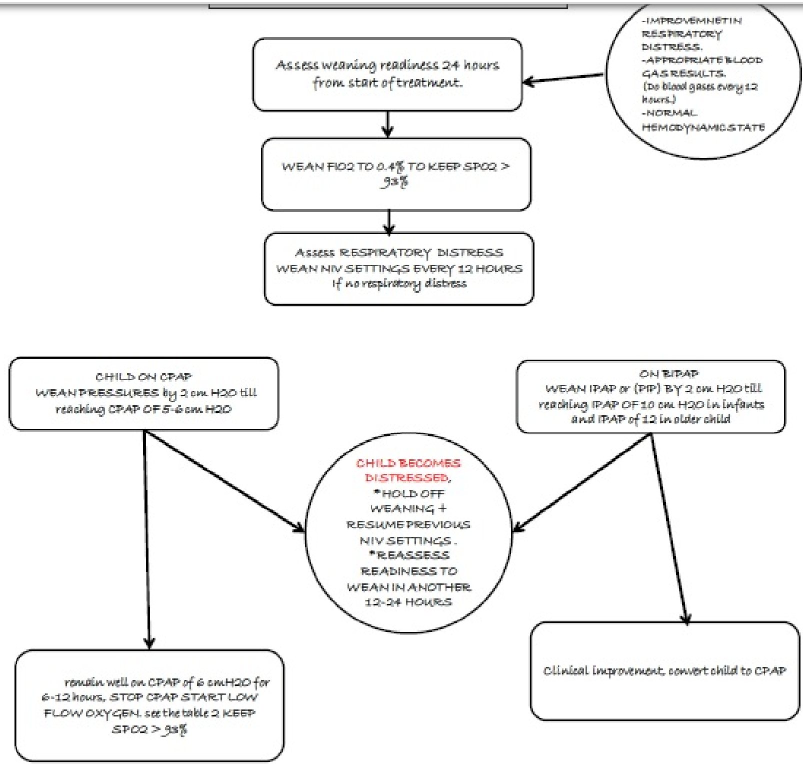
Clinical Improvement
-Continue monitoring for respiratory distress
-Consider feeding within 4 hours if stable clinically

No improvement in the work of breathing, spo2 and gas exchange

Consider invasive ventilation no longer than 4-6 hours post NIV

Ventilators used in NIV

NIV devices such as triology	Conventional ICU ventilator
Features: -single limb circuit -need to use vented mask and anti-asphyxial valves - PIP pressure on the ventilator include PEEP - Triology can be used in 5 kgs to adult 	Features: -Dual limb circuit -need non vented mask -PIP pressure on the ventilator is above PEEP -C2 Hamilton in Hd can be used in 2 kgs and above 



High Flow Nasal Cannula(HFNC) Initiation Algorithm

Indications	1. Moderate respiratory distress 2. Children with intolerance to the mask or interface Common diseases for HFNC use ❑ Bronchiolitis most common indication ❑ Other respiratory diseases causing respiratory distress such as pneumonia, asthma and heart failure
Contra-indications	❑ Significant respiratory distress(relative) / respiratory failure ❑ Hemodynamic instability ❑ Acute respiratory Distress Syndrome with significant hypoxemia ❑ Encephalopathy

Settings	Infant : Start Flow at 2l/kg/min, Fio2 60% Child : Start Flow 1 l/kg/min , Fio2 60%, max. flow 25 L/min ❑ Aim for saturation 93%-94% ❑ Titrate Fio2 to clinical response * Note that maximum flow is 25 L/min on HFNC devices in use in the department

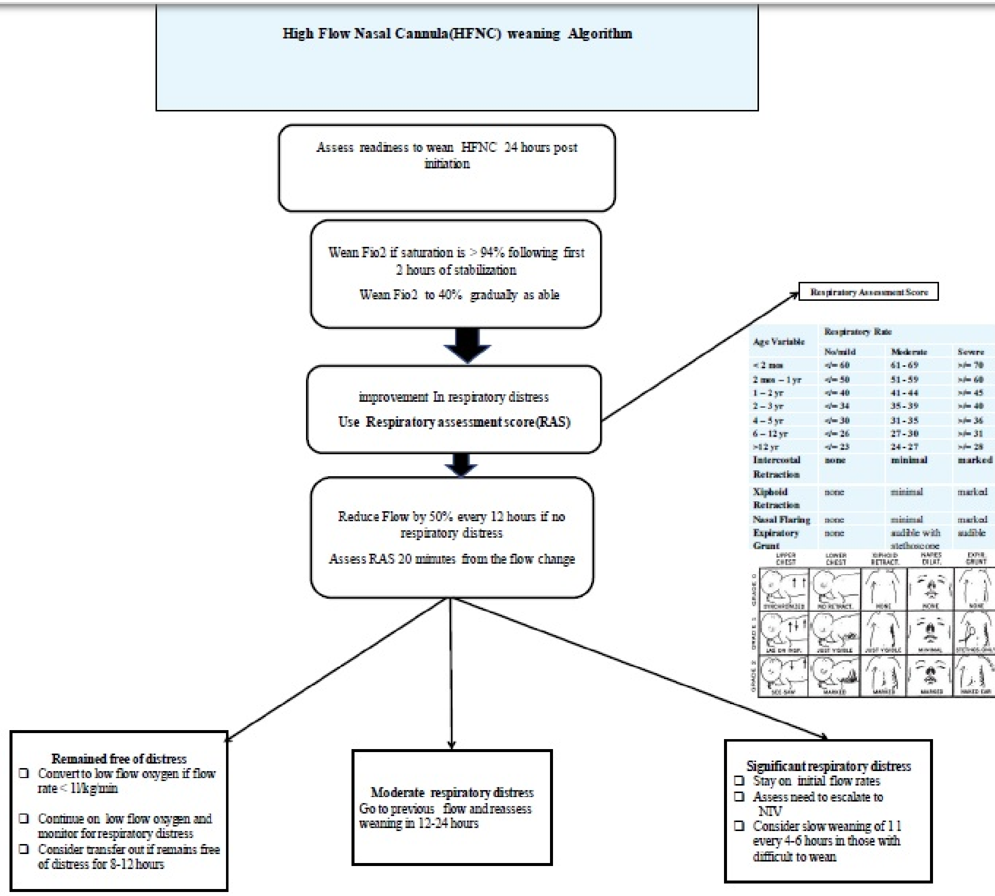
Monitoring	❑ Improvement in the work of breathing and accessory muscles use ❑ Improvement in oxygen saturation and blood gas ❑ Frequent clinical assessment required ❑ Most achieve clinical response within first 60 -90 minutes

Improvement in work of breathing, saturation and blood gas

❑ Continue clinical monitoring
❑ Consider feeding via NGT feeds within 4 hours if child is stable

No improvement in the work of breathing, saturation or blood gas

❑ Consult PICU
❑ Escalate treatment to invasive ventilation if in respiratory failure or non-invasive ventilation if not in respiratory failure
❑ Go to non-invasive ventilation algorithm



Age Variable	Respiratory Rate		
	No/mild	Moderate	Severe
< 2 mos	</= 60	61 - 69	>/= 70
2 mos – 1 yr	</= 50	51 - 59	>/= 60
1 – 2 yr	</= 40	41 - 44	>/= 45
2 – 3 yr	</= 34	35 - 39	>/= 40
4 – 5 yr	</= 30	31 - 35	>/= 36
6 – 12 yr	</= 26	27 - 30	>/= 31
>12 yr	</= 23	24 - 27	>/= 28
Intercostal Retraction	**none**	**minimal**	**marked**
Xiphoid Retraction	none	minimal	marked
Nasal Flaring	none	minimal	marked
Expiratory Grunt	none	audible with stethoscope	audible

Suggested low flow oxygen rates

Age	Flow Rate (L/min)
< 2 months	0.5
2 months – 2 years	1-1.5
3 – 4 years	2-3
> 4 years	No more than 4

References

1. Yañez LJ, Yunge M, Emilfork M, Lapadula M, Alcántara A, Fernández C, et al. A prospective, randomized, controlled trial of noninvasive ventilation in pediatric acute respiratory failure. Pediatr Crit Care Med 2008; 9:484–9. doi: 10.1097/PCC.0b013 e318184989f

2.Wolfler A, Calderini E, Iannella E, Conti G, Biban P, Dolcini A, et al. Evolution of noninvasive mechanical ventilation use: A cohort study among Italian PICUs. Pediatr Crit Care Med 2015; 16:418–27. doi: 10.1097/PCC.0000000000000387.

3.Essouri S, Chevret L, Durand P, Haas V, Fauroux B, Devictor D. Noninvasive positive pressure ventilation: Five years of experience in a pediatric intensive care unit. Pediatr Crit Care Med 2006; 7:329–34. doi: 10.1097/01.PCC.0000225089.21176.0B.

4. Milési C, Boubal M, Jacquot A, Baleine J, Durand S, Odena MP, et al. High-flow nasal cannula: Recommendations for daily practice in pediatrics. Ann Intensive Care 2014; 4:29. doi: 10.1186/ s13613-014-0029-5

5. Franklin D, Babl FE, Schlapbach LJ, Oakley E, Craig S, Neutze J, et al. A randomized trial of high-flow oxygen therapy in infants with bronchiolitis. N Engl J Med 2018; 378:1121–31. doi: 10.10 56/NEJMoa1714855

Written By: Dr.Kholoud Al Mukhaini
Checked By: PICU Team
Authorized by: KHOULA JULENDA AL SAID